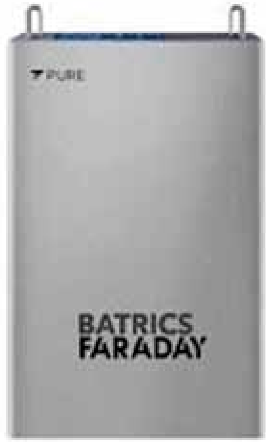
Tech News

Technology Updates

AI based system that resolves defects in lithium-ion batteries

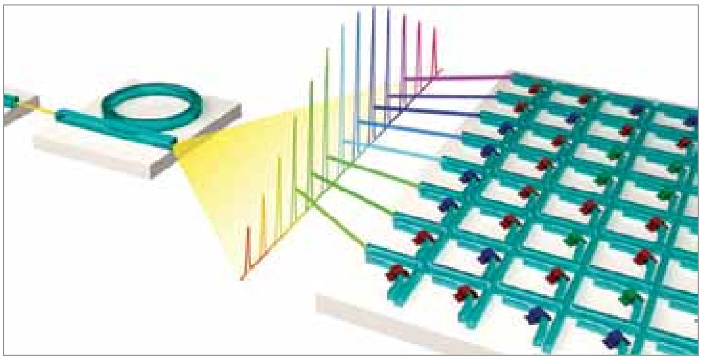
Electric vehicles (EVs) are gradually becoming popular as a viable eco-friendly mode of transportation. However, the problem of lithium battery repair presents a challenging task in the market as the supply chains are not well established. EV customers are reliant on the OEMs who, in turn, are reliant on battery manufacturers.

To solve this challenge, researchers at the IIT Hyderabad incubated startup Pure EV have designed artificial neural network (ANN) based algorithms for a system called BaTRics Faraday. The algorithms help identify battery cell defects and rectify the damage. The process is fully automated by the hardware and requires no manual intervention. The system is likely to be launched in the first quarter of 2021.



Light based chips advance machine learning

The computing demands for applications in artificial intelligence, such as pattern and speech recognition or for self-driving vehicles, often exceeds the capacities of conventional computer processors. To cope up with these tasks in an extremely efficient way, researchers at the University of Münster, along with an international team, are developing new approaches and process architectures with the help of photonic processors, which can rapidly process (up to 10^{12} - 10^{15} operations per second) data and information by means of light—something conventional electronic chips such as graphic cards or specialised hardware like Google's TPU (Tensor Processing Unit) are incapable of doing as these are based on electronic data transfer and are much slower.



Schematic representation of a processor for matrix multiplications which runs on light. Together with an optical frequency comb, the waveguide crossbar array permits highly parallel data processing (Credit: www.uni-muenster.de)

The use of larger neural networks allows more accurate forecasts and precise data analysis. For example, photonic processors support the evaluation of large quantities of data in medical diagnoses by generating high-resolution 3D data in special imaging methods. Further applications are in the fields of self-driving vehicles, which depend on fast, rapid evaluation of sensor data, and of IT infrastructures such as cloud computing that provide storage space, computing power, or applications software.

Enhanced optical-flow based control for drones

Flying insects heavily rely on optical flow for visual navigation and flight control. But when implemented in drones, they often run into problems that insects appear to have overcome. Now a team of researchers from TU Delft and the Westphalian University of Applied Sciences have developed an optical-flow based learning process that allows these small robots to estimate distances through the visual appearance (shape, colour, texture) of the objects in view, thus improving the navigation skills of drones.

Learning to see distances by means of visual appearance also leads to much faster, smoother landings. This technique is quite relevant to resource-constrained flying robots, especially when these operate in a rather confined environment, such as flying in greenhouses to monitor crops or keeping track of the stock in warehouses.